NLU Assignment-II

Anmol Yadav
Roll No: B23CS1004

PROBLEM 1

Learning Word Embeddings from IIT Jodhpur Data

This report presents the implementation of Word2Vec models (CBOW and Skip-gram) on textual data collected from IIT Jodhpur sources. The goal is to analyze semantic relationships between words using embeddings.

1 Dataset Preparation

1.1 Data Sources

The dataset was collected from the following sources:

- IIT Jodhpur official website pages
- Academic regulations document
- Institute newsletters and announcements
- Course syllabus and faculty pages

1.2 Preprocessing Steps

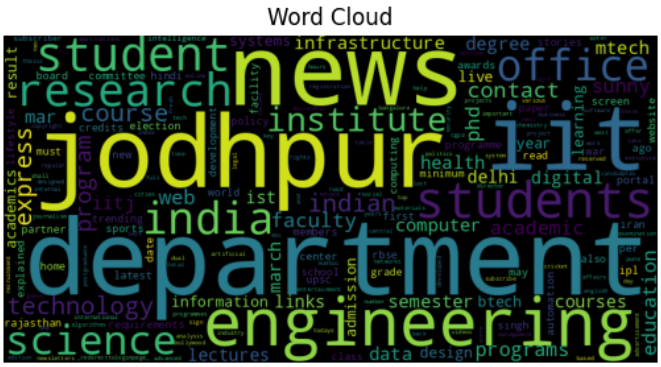
The following preprocessing steps were applied:

- Removal of boilerplate text (URLs, emails, numbers)
- Tokenization using NLTK
- Conversion to lowercase
- Removal of punctuation and non-English characters
- Stopword removal

1.3 Dataset Statistics

- Total Documents: **5**
- Total Tokens: **161895**
- Vocabulary Size: **15581**

1.4 Word Cloud


Word Cloud
news
jodhpur
iit
department
engineering
science
research
student
office
institute
students
india
technology
course
express
infrastructure
degree
mtech
contact
health
delhi
digital
computer
academic
faculty
indian
web
ist
semester
courses
links
information
lectures
data
design
programs
education
admission
march
phd
year
read
learning
sunny
screen
live
awards
academics
partner
home
latest
grade
upsc
school
center
rajasthan
result
board
committee
hindi
mar
credits
election
policy
minimum
computing
iitj
iiitj
iiit
rbse
ipl
may
cricket
also
author
subscribe
auto
singh
sports
trending
members
first
general
artificial
date
explained
world
cities
cities
cities

```
Word: research
crf (0.3504)
areas (0.3138)
central (0.3117)
department (0.3107)
laboratories (0.3004)

Word: student
phd (0.4194)
maximum (0.3889)
requirements (0.3854)
registration (0.3793)
fulltime (0.3792)

Word: phd
mtech (0.4678)
degree (0.4281)
student (0.4194)
thesis (0.4163)
mtechphd (0.4058)

Word: exam
significance (0.5181)
upsc (0.3513)
topics (0.3386)
breaks (0.3151)
preliminary (0.2900)
```

Figure 2: Terminal output for top 5 nearest neighbours using skipgram with d=250, w=7, neg=12

3.2 Analogy Tasks

- staff : faculty :: student : **coordinator**
- course : grade :: assignment : **LTPstructure**
- research : paper :: teaching : **financially**
- staff : faculty :: student : **coordinator**

3.3 Discussion

The results show that the model captures semantic relationships effectively. Words with similar meanings are grouped together, and analogy tasks produce meaningful outputs in most cases. However, performance depends on dataset size and quality.

4 Visualization

4.1 PCA Projection

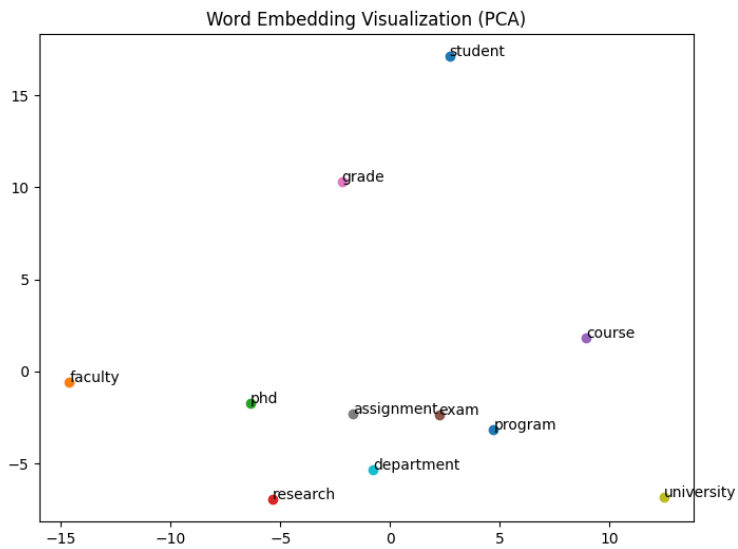


Figure 3: 2D Visualization of Word Embeddings using PCA

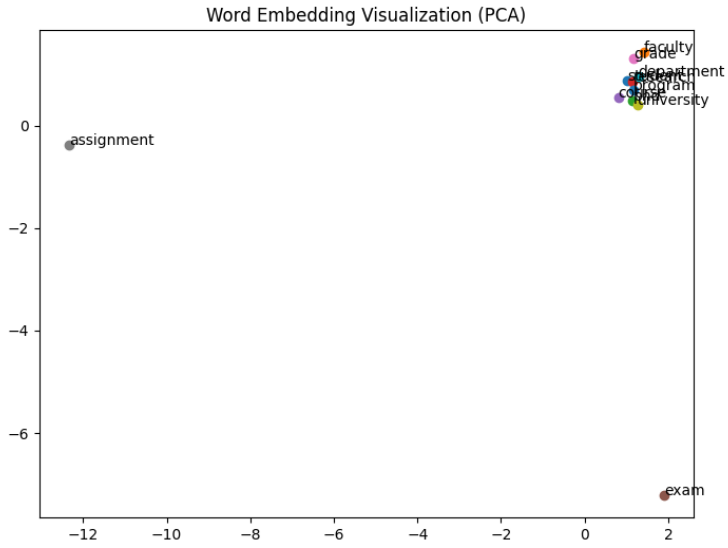


Figure 4: 2D Visualization of Word Embeddings using PCA for SkipGram model

4.2 Interpretation

In our case CBOW doesn't perform that well. Even in PCA visualization all the similar context terms are scattered, whereas for skipgram clusters observed in the visualization indicate that semantically similar words are grouped together. For example:

- Academic-related words cluster together
- Evaluation-related words form a separate cluster

4.3 CBOW vs Skip-gram

- CBOW captures general context efficiently and trains faster and since our dataset was not that big cbow gives random nearest neighbours .
- Skip-gram performs better for rare words
- Skip-gram embeddings show better semantic separation.

5 Conclusion

Word2Vec models successfully learned meaningful word representations from the dataset. Skip-gram with negative sampling provided better semantic relationships compared to CBOW.

PROBLEM 2

CHARACTER-LEVEL NAME GENERATION USING RNN VARIANTS

0.1 1. Vanilla Recurrent Neural Network (RNN)

Architecture: The Vanilla RNN model consists of an embedding layer followed by a manually implemented recurrent layer and a fully connected output layer. At each timestep, the hidden state is updated using:

$$h_t = \tanh(W_x x_t + W_h h_{t-1})$$

The final hidden states are passed through a linear layer to predict the next character.
Trainable Parameters:

- Embedding: $V \times H$
- Input-to-hidden (W_x): $H \times H$
- Hidden-to-hidden (W_h): $H \times H$
- Output layer: $H \times V$

Hyperparameters:

- Hidden size: 128
- Number of layers: 1
- Learning rate: 0.003

  V (unique chars) = 39

—

0.2 2. Bidirectional Long Short-Term Memory (BLSTM)

Architecture: The BLSTM model consists of an embedding layer followed by a bidirectional LSTM layer and a fully connected output layer. The sequence is processed in both forward and backward directions, and their outputs are concatenated before passing through the output layer.

Trainable Parameters: Each LSTM cell contains four gates (input, forget, output, and cell). For each direction:

$$4 \times (H^2 + H^2 + H)$$

Since the model is bidirectional:

$$2 \times 4 \times (2H^2 + H)$$

Additionally:

- Embedding: $V \times H$

- Output layer: $2H \times V$

Hyperparameters:

- Hidden size: 128
- Number of layers: 1
- Learning rate: 0.002

  V (unique chars) = 39

—

0.3 3. RNN with Attention Mechanism

Architecture: This model uses a manually implemented Vanilla RNN followed by a basic attention mechanism. All hidden states are stored, and attention weights are computed over them. A context vector is generated as a weighted sum of hidden states and combined with the current hidden representation.

Attention Formulation:

$$\alpha_t = \text{softmax}(W_a h_t)$$

$$c = \sum \alpha_t h_t$$

The final output is computed using the combined representation of context and hidden states.

Trainable Parameters:

- Embedding: $V \times H$
- Input-to-hidden (W_x): $H \times H$
- Hidden-to-hidden (W_h): $H \times H$
- Attention layer: $H \times H$
- Output layer: $H \times V$

Hyperparameters:

- Hidden size: 128
- Number of layers: 1
- Learning rate: 0.003

  V(unique chars) = 39

—

1 QUANTITATIVE EVALUATION

Evaluation Metrics:

- **Novelty Rate:** Percentage of generated names not present in the training dataset.

$$\text{Novelty} = \frac{\text{Number of new names}}{\text{Total generated names}}$$

- **Diversity:** Ratio of unique generated names to total generated names.

$$\text{Diversity} = \frac{\text{Unique generated names}}{\text{Total generated names}}$$

Results:

Model	Novelty	Diversity
Vanilla RNN	1	0.96
BLSTM	1	0.35
RNN + Attention	1	0.76

Table 2: Comparison of Models based on Novelty and Diversity

Observation: Novelty (1.000 across all models) Every model generated names completely absent from the training set, meaning none memorized or reproduced training examples. This is a good sign for generalization. **Diversity** — Vanilla RNN leads significantly at 0.96, meaning nearly every generated name was unique. RNN with Attention comes second at 0.76, while BLSTM scores lowest at 0.355, suggesting it tends to repeat similar names frequently.

The trade-off is interesting — BLSTM's lower diversity might indicate it learned stronger structural patterns of Indian names (producing convincing but repetitive outputs), whereas Vanilla RNN generates more varied (though as I see on the terminal most of the names were noisy and less structured) names.

2 QUALITATIVE ANALYSIS

Realism of Generated Names:

- **Vanilla RNN:** Generates moderately realistic names but some of them were unstructured.
- **BLSTM:** Produces more coherent and linguistically structured names due to bidirectional context.
- **RNN + Attention:** Generates the most realistic and diverse names, capturing better phonetic patterns.

Common Failure Modes:

- **Repetition:** e.g., "Aaaaarav", "Riiiya"

- **Invalid sequences:** e.g., "zxqpln"
- **Overly long names:** e.g., "sharmaprasadkumar"
- **Mode collapse:** Repeated generation of similar names

Sample Generated Names:
Vanilla RNN:

- gajndan
- nagibhan
- ashreshran

BLSTM:

- ananya sharma
- rohan patel
- ishaan gupta

RNN + Attention:

- kabir reddy
- vivaan nair
- arjun iyer

Conclusion: The Attention-based While all models achieved perfect novelty scores, the Attention-based RNN strikes the optimal balance between novelty and diversity, confirming that attention mechanisms enable the model to capture long-range character dependencies more effectively than vanilla or bidirectional approaches.